

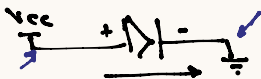
* H.W Component

1) Io → Control 2) Digital H.W Component

- ① led
- ② buzzer
- ③ segment
- ④ button
- ⑤ keypad
- ⑥ LCD
- ⑦ Digital sensor

* H.W Component

① led → lighting emitting diode → output



* Connection type

Source Connection



led on

led off

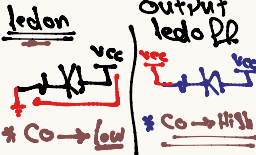
* Co → Direction
Output

* Co → output
High value

Co → Direction
output

Co → output value
low

Sink Connection



* Co → low

* Co → High

* Source Connection

- ① Mc / 9nd
- ② led on → High / led off → Low

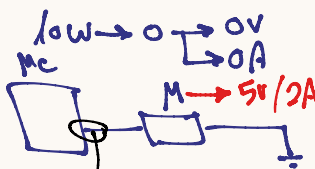
* Sink Connection

- ① Mc / Vcc
- ② led on → Low / led off → High

* How is better [Source [Sink]

- For education → Source
- For Prediction → Based on load

ex: MCU → High → I → Vcc / 40mA



* Source Connection Problem: if you have load take more than provide (40mA)

* MCU → better Pin

Note

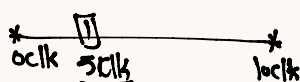
- Embedded system have multi state code system
- ① super loop system
- ② For Background system
- ③ RTOS system

* Super loop system

→ all logic application running in while(1)

→ High Determinism | Low Responstimes

```
while(1)
{
    Button();
    led();
    servos();
    lcd();
    Aadc();
}
```



* waiting type

① Blocking wait

- Project blocked.
- for loop → nothing
- delay_ms(1000);

② timer

③ os delay

* for 1

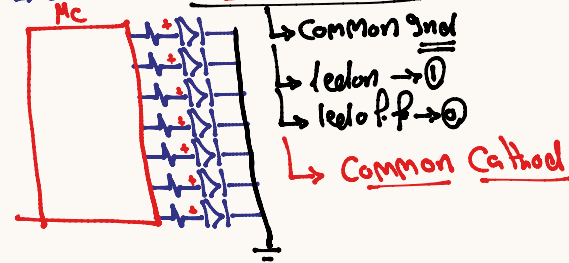
→ toggle led every 500ms

→ group pin

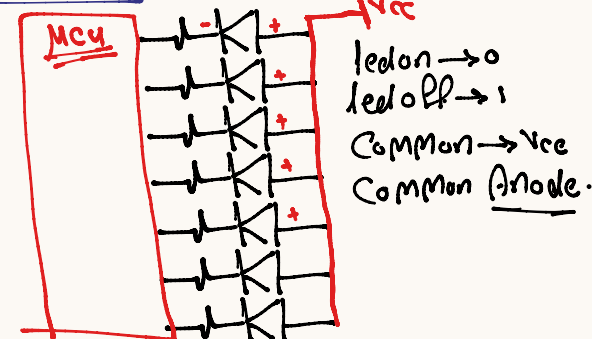
Source Connection Sink Connection

* if you have 7-led need to control

→ Connection → source Connection



→ Connection → Sink Connection



* Display Component

- ① led → Notification
- ② segment → 7-segment → Display Number
- ③ screen

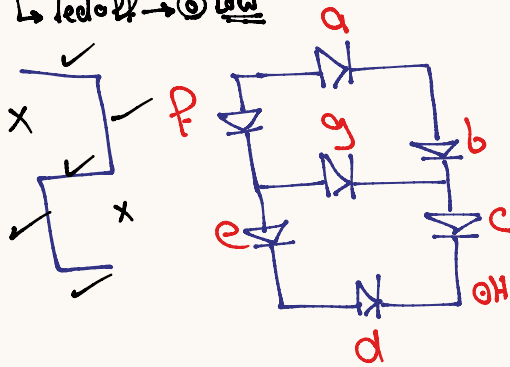
* segment → 7-segment → 7-lead connect together
↓ size

Common Cathode

- Common → Signal
- source Connection
- led on → ① High
- led off → ② Low

Common Anode

- Common Vcc
- Sink Connection
- led on → low
- led off → High



* Case 1 → team 1

→ Cathode | H g f e d c b a
7 6 5 4 3 2 1 0
0 1 0 1 1 0 1 1
0x5B

0 1 2 3 4 5 6 7 8 9

* Case 2 → team 2

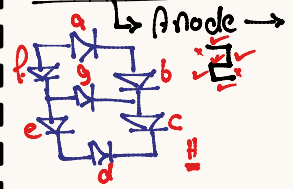
→ Cathode | a b c d e f g h
7 6 5 4 3 2 1 0
0 x f 2 1 1 1 1 0 0 1 0
0 x D A 1 1 0 1 1 0 1 0

* Case 3 → team 3

Anode → on → Low (0)
off → High (1)

→ Anode | g f e d c b a
7 6 5 4 3 2 1 0
1 0 1 0 0 1 0 0
0xA4

* team 4 → Case 4



on → 0
off → 1
a b c d e f g h
7 6 5 4 3 2 1 0
0 0 1 0 0 1 0 1
0x25

to control on 7 segment need # of pin
8 → group

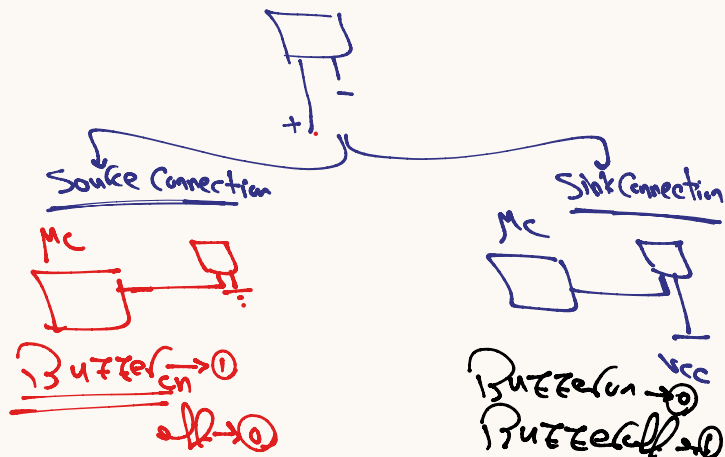
* Per group → output

group → Drive
Write.

* output H.W Component

→ Buzzer → sound

We control on Buzzer like led



H.W Component Input

→ Switch → Connect / Disconnect between two point

pressed → Connect
not pressed → dis Connect.

→ type → ① Mechanical switch

need Human Power to change from state to another

→ electrical switch

* need electricity to change from state to another

* Mechanical Switch

↳ ① Normal Button

it Button save the last state if the human power loss

↳ ② Push Button

it Button Return to default state if the human power loss.



* De bouncing

Push Button → Spring →

After Remove human power → spring still vibrate → can be connect switch

still for milli seconds.

MC → speed → 8 MIPS

H.W Solving

- * RC - circuit
- * SR - latch

SW Solving

* delay

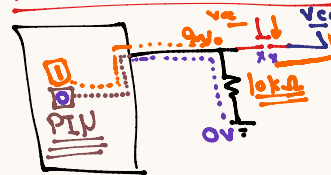
→ Read Button

↳ Pressed → delay

* timer Read Button

* Button Connection

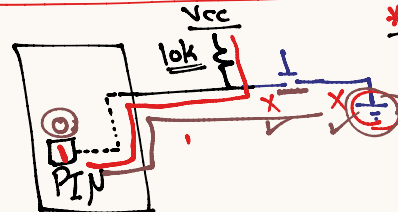
① External Pull down



② Direction (Input)

↳ Not Pressed
↳ PIN → 0
↳ Pressed
↳ PIN → 1

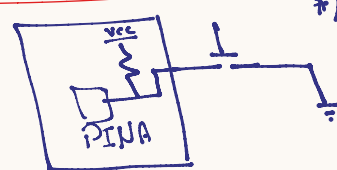
② External Pull up



Direction → Input

* Not Pressed = 1
Pressed = 0

③ Internal Pull up



* Pressed → 0
* Not Pressed → 1

* How Enable Internal Pull up

↳ Pin → ① Direction → Input

DDR → 0

Direction select

② Output → High

PORT → 1

Write Pin

Write Pin

if Pin output
Direction

Provide
High/Low → outside
Pin

if Pin Input

Enable/Disable
Internal Pull up

